Control System: Phases 0 and 1

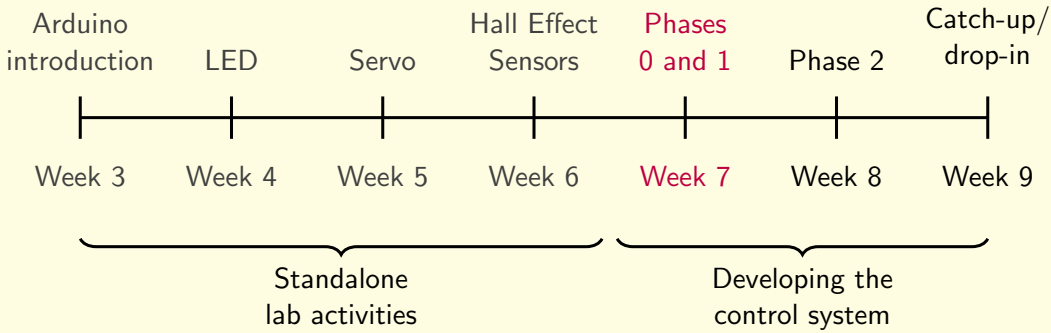
Dr Max Champneys

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University of Sheffield, UK

MAC 233: **11/11/2026**



MAC 233: LABS OVERVIEW

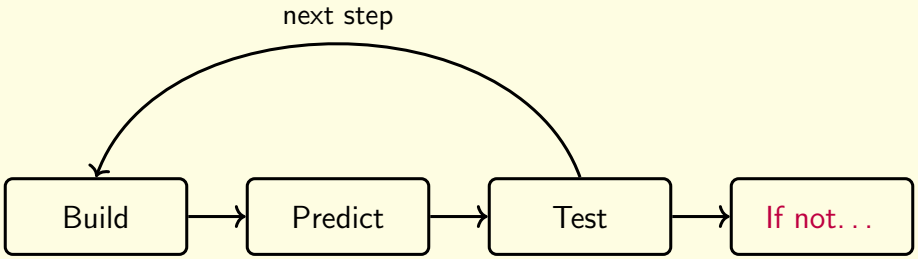


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INTEGRATION HELL

Build everything, then test it: if it doesn't work, where is the fault?

Build one part, then test it: the fault is in what you just added.



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TODAY

Phase 0: Software. Read through and *understand* `control-system.ino` then answer the check questions in your groups.

Phase 1: Electronics. Build each component in steps 1–5, testing each one before moving on. Steps 1–4 are on the bench (pass the magnet by hand to test the sensors); step 5 moves onto your mini rig.

Phase 2 Integration (next week): test the example controller on the mini-rig, then alter the sketch to make it better.

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CONTROL-SYSTEM.INO

We provide to you a complete sketch of a basic control system, `control-system.ino`

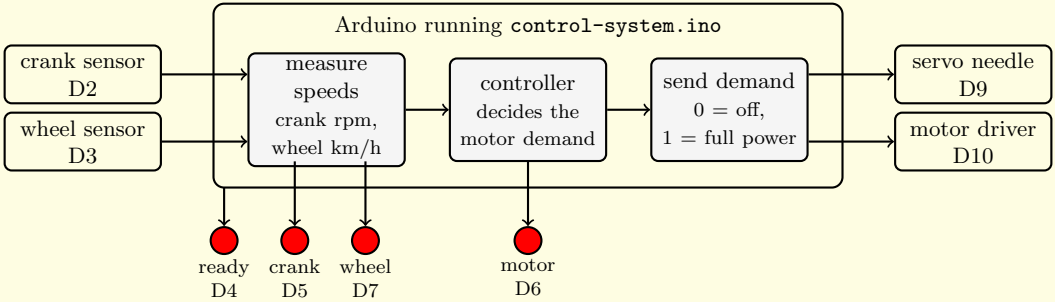
- reads hall effect sensors to compute speeds
- decides how much power to send to the motor
- outputs to LEDs and the serial montior

It works on both the mini rig (Autumn semester) and the big rigs (Spring Semester).

Upload it to the arduino once, at the start of phase 1. No need to change it until Phase 2.

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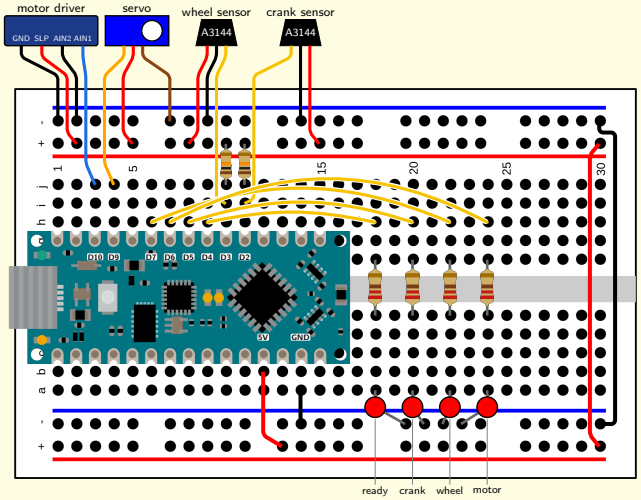
PHASE 0: SOFTWARE



Only help while the rider is pedalling, and never above 25 km/h.

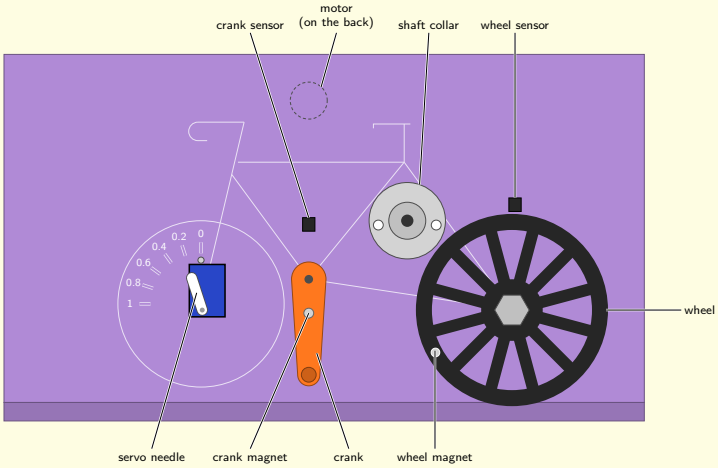
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PHASE 1: ELECTRONICS



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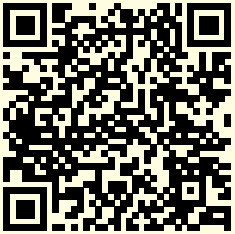
HARDWARE



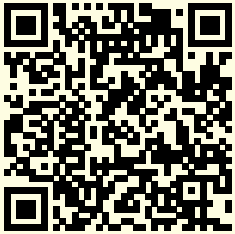
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HELP

Worksheet



Sketch



Nano Every pinout



Problems? Raise your hand and someone will come and help you.

Errors? Let me know and win some prizes!

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